

DataVortex :: Round 2 SQL Challenge

Human-Centric Logic & Architectural Query Explanations

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Target Database Tables: `posts` , `users` , `posts_corrupted` , `posts_cleaned`

Executive Overview: How We Approached the Queries

When writing SQL for high-stakes data engineering and competitive analytics, the biggest mistake is writing code that merely "compiles" without understanding the real-world domain. In the DataVortex Social Engine, the dataset came from an unstable stream where network retries, bit inversions, and asynchronous microservices created deliberate corruptions and subtle statistical quirks.

Every query in our suite was engineered with four human-first principles:

1. **Defensive Filtering:** Never let garbage data silently contaminate aggregate metrics (handling `NULL` , empty strings, and negative values explicitly).
 2. **Semantic Integrity:** When calculating "engagement," ensuring that missing likes don't inadvertently register as zero engagement, and using cleaned/imputed fields where appropriate (`COALESCE(likes_cleaned, likes)`).
 3. **Readable Modular CTEs:** Breaking complex multi-tier business questions into logical Common Table Expressions (`WITH ... AS`) so that another engineer or judge can instantly trace the mathematical flow.
 4. **Analytical Storytelling:** Looking past the raw numbers to understand what the output actually tells us about social algorithms, user behaviors, and platform dynamics.
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Part 1: Easy Level Challenges (E1 – E5)

E1 — Platform Popularity

1. Plain-English Objective

Find out which social media platform is the most popular in terms of raw volume (the highest number of published posts), while making sure we completely ignore records where the platform name is blank or missing.

2. The Thought Process & Data Traps

- In the raw dataset, approximately 15% of posts lost their platform tag during the intake crash, resulting in either SQL `NULL` or empty strings (`' '`).
- If you do a simple `GROUP BY platform` , your top bucket might literally be `NULL` or an empty row, giving a false answer.
- We must apply defensive filtering (`WHERE platform IS NOT NULL AND TRIM(platform) ≠ ''`), count the rows for each platform, sort from highest to lowest, and pick the single top result.

3. SQL Query

```
SELECT
    platform,
    COUNT(*) AS post_count
FROM posts
WHERE platform IS NOT NULL
    AND TRIM(platform) ≠ ''
GROUP BY platform
ORDER BY post_count DESC
LIMIT 1;
```

4. Query Output

platform	post_count
Facebook	2,074

5. Human Explanation of the Result

Facebook edges out the competition with 2,074 posts, closely followed by YouTube (2,071), Twitter (2,060), and Reddit (2,028). The post distribution across all five platforms is nearly identical (~20% each), proving that before the crash, the intake daemon was load-balancing posts evenly across platform queues.

E2 — Most Engaged Posts

1. Plain-English Objective

Identify the 10 absolute "mega-hit" viral posts across the entire platform based on total community interaction—defined as the sum of likes, shares, and comments. Exclude posts where like counts are missing.

2. The Thought Process & Data Traps

- In SQL, evaluating `NULL + shares + comments` yields `NULL`. If a post had 2,000 shares and 1,000 comments but its likes were `NULL`, standard addition would completely erase that post from the ranking.
- The prompt strictly instructs us to *ignore posts where likes are missing*. Therefore, an explicit `WHERE likes IS NOT NULL` is required.
- We add all three interaction types `(likes + shares + comments) AS total_engagement`, sort in descending order, and take the top 10.

3. SQL Query

```
SELECT
    post_id,
    platform,
    likes,
    shares,
    comments,
    (likes + shares + comments) AS total_engagement
FROM posts
WHERE likes IS NOT NULL
ORDER BY total_engagement DESC
LIMIT 10;
```

4. Query Output

post_id	platform	likes	shares	comments	total_engagement
ycjj5zzt7mvx	Instagram	4,983	1,919	991	7,893
wo7py9aljg3t	Reddit	4,864	1,981	948	7,793
gmoeib832zbs	Facebook	4,902	1,880	982	7,764
5kvuyvf38nqx	YouTube	4,923	1,971	861	7,755
pvfl3d8hj7jd	Instagram	4,989	1,840	909	7,738
tdgjyylpua20	UNKNOWN	4,979	1,932	812	7,723
tne7s3o4l4wd	Instagram	4,931	1,903	878	7,712
a1kiwl618kzy	Facebook	4,811	1,952	920	7,683
fp89q1ickn9w	Twitter	4,740	1,933	955	7,628
5n161ir5hhhr	YouTube	4,751	1,981	878	7,610

5. Human Explanation of the Result

The ceiling for total engagement tops out around 7,900. Notice how the individual metrics are capped by the platform's generation rules: likes max out near 5,000, shares near 2,000, and comments near 1,000. Instagram claims the #1 spot, but high virality occurs across all platforms.

E3 — Average Engagement by Platform

1. Plain-English Objective

Calculate the typical (average) likes, shares, comments, and total interaction generated per post on each social network, and determine which platform delivers the highest overall return for creators.

2. The Thought Process & Data Traps

- If we simply compute `AVG(likes + shares + comments)` on raw corrupted data, rows with `NULL` likes will be completely ignored by SQL's `AVG()`, skewing the sample.
- By utilizing `COALESCE(likes_cleaned, likes)`, we leverage the machine-learning-restored likes count (CatBoost + Predictive Mean Matching) or fallback to raw likes without distorting the sample.
- We round each average to 2 decimal places for clean reporting and sort by `avg_total_engagement DESC`.

3. SQL Query

```
SELECT
    platform,
    ROUND(AVG(COALESCE(likes_cleaned, likes)), 2) AS avg_likes,
    ROUND(AVG(shares), 2) AS avg_shares,
    ROUND(AVG(comments), 2) AS avg_comments,
    ROUND(AVG(COALESCE(likes_cleaned, likes, 0) + shares + comments), 2) AS avg_total_engagement
FROM posts
WHERE platform IS NOT NULL
GROUP BY platform
ORDER BY avg_total_engagement DESC;
```

4. Query Output

platform	avg_likes	avg_shares	avg_comments	avg_total_engagement
Instagram	2,502.99	1,040.84	499.80	4,043.63
YouTube	2,502.65	1,011.83	504.38	4,018.86
Facebook	2,508.24	984.17	506.94	3,999.35
Reddit	2,465.48	1,002.23	511.18	3,978.89
UNKNOWN	2,481.21	998.61	496.54	3,976.35
Twitter	2,448.89	1,005.39	506.13	3,960.41

5. Human Explanation of the Result

Instagram emerges as #1 in total engagement, generating an average of 4,043.63 interactions per post, driven by superior share volume (1,040.84). Interestingly, **Reddit wins on discussion depth**, yielding the highest average comment count (511.18). The tight clustering across all platforms (~4,000 interactions) proves the recommendation algorithm maintains consistent engagement baselines across networks.

E4 — Highly Shared but Poorly Liked

1. Plain-English Objective

Find the "viral anomalies": posts that people shared aggressively (over 1,500 shares) but almost nobody liked (under 500 likes).

2. The Thought Process & Data Traps

- In social media forensics, high shares with low likes usually indicate:
 1. Highly controversial or offensive content ("Look at what this person just posted!").
 2. Public service warnings or breaking alerts where hitting "like" feels inappropriate.
 3. Algorithmic amplification bot behavior.
- Trap: Raw likes contained negative numbers (e.g. `-4812`). If we did `likes < 500` without filtering out negatives or cleaning them, an inverted like count of `-4812` (which was really `+4812`) would falsely show up here! We ensure `likes ≥ 0` (or utilize cleaned likes) to avoid capturing sign-flipped rows.

3. SQL Query

```
SELECT
  post_id,
  platform,
  likes,
  shares,
  comments
FROM posts
WHERE shares > 1500
  AND likes < 500
  AND likes ≥ 0
ORDER BY shares DESC
LIMIT 15;
```

4. Query Output (Top 5 Sample)

post_id	platform	likes	shares	comments
<code>euvr0r10wrj6</code>	Facebook	453	2,000	408
<code>oiszojqm6qnn</code>	Instagram	390	1,999	858
<code>f2e5kdfldedz</code>	UNKNOWN	447	1,997	641
<code>mgv7p46wzpek</code>	Reddit	252	1,993	971
<code>lit2hyqg0v0l</code>	Facebook	14	1,990	769

5. Human Explanation of the Result

Look at post `lit2hyqg0v0l` on Facebook: **only 14 likes, but 1,990 shares and 769 comments!** That represents a share-to-like ratio of over 142:1. This is a classic hallmark of high-friction social discourse.

E5 — Users With Large Audiences

1. Plain-English Objective

Extract the platform's "macro-influencers"—every user who has built an audience of more than 40,000 followers—along with their location and primary language.

2. The Thought Process & Data Traps

- This is a straightforward user-demographic filter on `users.follower_count > 40000`.
- We order descending by follower count so the most prominent creators appear at the top.

3. SQL Query

```
SELECT
    user_id,
    location,
    language,
    follower_count
FROM users
WHERE follower_count > 40000
ORDER BY follower_count DESC
LIMIT 15;
```

4. Query Output (Top 5 Sample)

user_id	location	language	follower_count
user_3o7w66o2	Berlin, Germany	ja	49,944
user_u98jwp3f	Chicago, USA	zh	49,936
user_usts5yuo	Shanghai, China	ja	49,933
user_d4eat3v3	Tokyo, Japan	de	49,914
user_siuvpkza	Chicago, USA	en	49,905

5. Human Explanation of the Result

Follower counts in this platform top out just below 50,000. Notice the cosmopolitan demographic spread: creators in Berlin tweeting in Japanese, Chicago creators posting in Chinese, and Tokyo creators posting in German.

Part 2: Medium Level Challenges (M1 – M5)

M1 — Which Locations Generate the Most Engagement?

1. Plain-English Objective

Join user demographic profiles with post activity to determine which global cities/locations generate the largest total volume of social interaction.

2. The Thought Process & Data Traps

- Posts live in `posts`, but geographic location lives in `users`. We must execute an `INNER JOIN` on `user_id`.
- We group by `u.location`, count the posts from creators in that city, and sum their total engagement.
- Using `COALESCE(p.likes_cleaned, p.likes, 0)` guarantees that a missing like doesn't turn the sum into `NULL`.

3. SQL Query

```
SELECT
    u.location,
    COUNT(p.post_id) AS post_count,
    SUM(COALESCE(p.likes_cleaned, p.likes, 0) + p.shares + p.comments) AS total_engagement
FROM users u
JOIN posts p ON u.user_id = p.user_id
GROUP BY u.location
ORDER BY total_engagement DESC
LIMIT 15;
```

4. Query Output (Top 5 Sample)

location	post_count	total_engagement
Los Angeles, USA	459	1,850,886
Munich, Germany	452	1,802,490
Shanghai, China	451	1,795,875
Barcelona, Spain	439	1,769,556
Melbourne, Australia	421	1,706,510

5. Human Explanation of the Result

Los Angeles leads the world with 1.85 million interactions across 459 posts, closely trailed by Munich and Shanghai. The geographic engagement is heavily correlated with post volume—cities with more active posting naturally accumulate more aggregate interactions.

M2 — Do High Follower Users Get More Engagement?

1. Plain-English Objective

Test the common industry assumption: *Does having more followers actually get you more engagement per post?* Split users into two equal halves (high followers $\geq 25,000$ vs. low followers $< 25,000$) and compare their average engagement.

2. The Thought Process & Data Traps

- We use a `CASE` statement to divide users into two distinct cohorts:
- `High follower users ( $\geq 25,000$ )`
- `Low follower users ( $< 25,000$ )`
- We join `users` with `posts` and calculate `AVG(total_engagement)` per post for each cohort.

3. SQL Query

```
SELECT
  CASE
    WHEN u.follower_count ≥ 25000 THEN 'High follower users (≥ 25,000)'
    ELSE 'Low follower users (< 25,000)'
  END AS user_group,
  COUNT(p.post_id) AS total_posts,
  ROUND(AVG(COALESCE(p.likes_cleaned, p.likes, 0) + p.shares + p.comments), 2) AS avg_engagement_p
FROM users u
JOIN posts p ON u.user_id = p.user_id
GROUP BY
  CASE
    WHEN u.follower_count ≥ 25000 THEN 'High follower users (≥ 25,000)'
    ELSE 'Low follower users (< 25,000)'
  END;
```

4. Query Output

user_group	total_posts	avg_engagement_per_post
High follower users (\$≥\$ 25,000)	5,925	3,997.41
Low follower users (\$<\$ 25,000)	6,075	3,995.67

5. Human Explanation of the Result (The Big Revelation!)

The difference is a mere 1.74 interactions per post (3,997.41 vs. 3,995.67)!

This is one of the most crucial findings in the entire DataVortex project: **follower count has virtually ZERO impact on engagement per post**. The platform's algorithm distributes content democratically based on content quality/relevance rather than legacy audience size.

M3 — Most Active Users

1. Plain-English Objective

Identify the platform's "power creators"—the top 10 individual users who have published the most posts—and inspect their follower count and overall engagement.

2. The Thought Process & Data Traps

- We group by user details (`u.user_id`, `u.follower_count`, `u.location`), count posts using `COUNT(p.post_id)` , and sum total engagement.
- We sort primarily by `post_count DESC` and secondarily by `total_engagement DESC` .

3. SQL Query

```
SELECT
    u.user_id,
    u.follower_count,
    u.location,
    COUNT(p.post_id) AS post_count,
    SUM(COALESCE(p.likes_cleaned, p.likes, 0) + p.shares + p.comments) AS total_engagement
FROM users u
JOIN posts p ON u.user_id = p.user_id
GROUP BY u.user_id, u.follower_count, u.location
ORDER BY post_count DESC, total_engagement DESC
LIMIT 10;
```

4. Query Output

user_id	follower_count	location	post_count	total_engagement
user_zqv2vrf5	13,531	Vancouver, Canada	22	85,967
user_nfo3ih5u	40,429	Barcelona, Spain	19	79,421
user_q3vs0uj7	32,714	Manchester, UK	18	67,142
user_8wk0j5ae	36,943	Los Angeles, USA	18	65,496
user_n0ok02rt	2,531	Dubai, UAE	18	63,716
user_0irp4abu	44,206	Toronto, Canada	17	73,245
user_ujllj1n7	49,855	Chicago, USA	16	72,227
user_uerv85na	1,824	Rome, Italy	16	71,836
user_hdas0iau	1,620	Rio de Janeiro, Brazil	16	66,500
user_i1e1kek5	46,765	Vancouver, Canada	16	66,244

5. Human Explanation of the Result

user_zqv2vrf5 in Vancouver leads the platform with 22 published posts and 85.9k engagement. Note that creators with fewer than 2,000 followers (user_uerv85na with 1,824 followers; user_hdas0iau with 1,620 followers) are posting just as prolifically and generating over 66,000+ interactions!

M4 — Platform Behaviour by High Follower Users

1. Plain-English Objective

When macro-creators (users with $\geq 30,000$ followers) post, which specific social media platform rewards them with the highest average engagement per post?

2. The Thought Process & Data Traps

- We filter for macro-influencers in the WHERE clause: `u.follower_count ≥ 30000`.

- We group by `p.platform`, count how many posts macro-creators published on each platform, and calculate their average engagement.

3. SQL Query

```
SELECT
    p.platform,
    COUNT(p.post_id) AS post_count,
    ROUND(AVG(COALESCE(p.likes_cleaned, p.likes, 0) + p.shares + p.comments), 2) AS avg_engagement_p
FROM users u
JOIN posts p ON u.user_id = p.user_id
WHERE u.follower_count ≥ 30000
    AND p.platform IS NOT NULL
GROUP BY p.platform
ORDER BY avg_engagement_per_post DESC;
```

4. Query Output

platform	post_count	avg_engagement_per_post
Instagram	809	4,109.61
UNKNOWN	681	4,001.45
Twitter	785	3,998.17
Facebook	827	3,981.08
Reddit	799	3,978.70
YouTube	846	3,953.80

5. Human Explanation of the Result

Instagram provides the highest return for macro-influencers, averaging 4,109.61 interactions per post—over 150 interactions higher than YouTube (3,953.80). Visual and aesthetic media formats on Instagram offer greater shareability for established influencers.

M5 — Detect Suspicious Engagement

1. Plain-English Objective

Detect potential bot manipulation or viral anomalies by finding posts where the number of shares alone exceeds the total number of likes and comments combined.

2. The Thought Process & Data Traps

- Normally, likes are easy to give (1 tap), comments take effort, and shares take conscious endorsement. A healthy post usually has `likes > shares`.
- If `shares > (likes + comments)`, it represents unusual viral propagation, coordinated bot retweeting, or extreme controversy.
- We calculate `share_excess = shares - (likes + comments)` and order by `shares DESC`.

3. SQL Query

```
SELECT
    post_id,
    platform,
    likes,
    shares,
    comments,
    (COALESCE(likes_cleaned, likes, 0) + comments) AS likes_plus_comments,
    (shares - (COALESCE(likes_cleaned, likes, 0) + comments)) AS share_excess
FROM posts
WHERE shares > (COALESCE(likes_cleaned, likes, 0) + comments)
ORDER BY shares DESC
LIMIT 20;
```

4. Query Output (Top 5 Sample)

post_id	platform	likes	shares	comments	likes_plus_comments	share_excess
euivr0r10wrj6	Facebook	453	2,000	408	861	+1,139
oiszojqm6qnn	Instagram	390	1,999	858	1,248	+751
2xcg9ld7du67	Twitter	1,425	1,999	440	1,865	+134
qq86lkjrfzlt	YouTube	897	1,999	850	1,747	+252
u1aa801qvxeu	UNKNOWN	162	1,992	306	468	+1,524

5. Human Explanation of the Result

On post `u1aa801qvxeu`, shares reached 1,992 while likes + comments combined only amounted to 468 (a surplus of +1,524 shares!). In a trust & safety audit, these 20 posts represent top-priority candidates for coordinated inauthentic behavior investigation.

Part 3: Hard Level Challenges (H1 – H6)

H1 — Find Users With Abnormally High Engagement

1. Plain-English Objective

Identify creators who are massively outperforming the platform: users whose total engagement is **more than twice (\$> 2\times\$) the overall platform average across all users**.

2. The Thought Process & Data Traps

- This requires a multi-stage aggregation:
 - Stage 1 (CTE `UserTotalStats`):** Calculate the total and average engagement for *every single user*.
 - Stage 2 (CTE `OverallUserAvg`):** Calculate the benchmark mean of all user totals across the database.
 - Stage 3:** Join the two CTEs and filter for users where `u.total_engagement > 2 * o.avg_user_total`.

3. SQL Query

```
WITH UserTotalStats AS (  
    SELECT  
        u.user_id,  
        u.location,  
        u.follower_count,  
        COUNT(p.post_id) AS post_count,  
        ROUND(AVG(COALESCE(p.likes_cleaned, p.likes, 0) + p.shares + p.comments), 2) AS avg_engagement,  
        SUM(COALESCE(p.likes_cleaned, p.likes, 0) + p.shares + p.comments) AS total_engagement  
    FROM users u  
    JOIN posts p ON u.user_id = p.user_id  
    GROUP BY u.user_id, u.location, u.follower_count  
)  
OverallUserAvg AS (  
    SELECT AVG(total_engagement) AS avg_user_total  
    FROM UserTotalStats  
)  
SELECT  
    u.user_id,  
    u.location,  
    u.follower_count,  
    u.post_count,  
    u.avg_engagement,  
    u.total_engagement  
FROM UserTotalStats u  
CROSS JOIN OverallUserAvg o  
WHERE u.total_engagement > 2 * o.avg_user_total  
ORDER BY u.total_engagement DESC  
LIMIT 10;
```

4. Query Output

user_id	location	follower_count	post_count	avg_engagement	total_engagement
user_zqv2vrf5	Vancouver, Canada	13,531	22	3,907.59	85,967
user_nfo3ih5u	Barcelona, Spain	40,429	19	4,180.05	79,421
user_0irp4abu	Toronto, Canada	44,206	17	4,308.53	73,245
user_ujllj1n7	Chicago, USA	49,855	16	4,514.19	72,227
user_uerv85na	Rome, Italy	1,824	16	4,489.75	71,836
user_2ytzut7i	Mumbai, India	17,064	14	4,906.57	68,692
user_q3vs0uj7	Manchester, UK	32,714	18	3,730.11	67,142
user_s87enj6b	Mexico City, Mexico	43,476	14	4,776.07	66,865
user_hdas0iau	Rio de Janeiro, Brazil	1,620	16	4,156.25	66,500
user_i1e1kek5	Vancouver, Canada	46,765	16	4,140.25	66,244

5. Human Explanation of the Result

The overall average total engagement across all users is ~31,980. The 10 creators above achieved more than double that threshold (>64,000 engagement), combining high posting volume (14 to 22 posts) with stellar per-post engagement (up to 4,906 interactions per post for `user_2ytzut7i` in Mumbai).

H2 — Rank Users Within Their Location

1. Plain-English Objective

For every global city in the dataset, rank its local creators by total engagement and return the top 3 podium winners (1st, 2nd, and 3rd place) for each location.

2. The Thought Process & Data Traps

- If you do a simple `GROUP BY` with `LIMIT 3`, you only get 3 rows total for the entire database.
- To rank *within* each city, we must use an SQL Window Function:
`DENSE_RANK() OVER (PARTITION BY location ORDER BY total_engagement DESC)`.
- `PARTITION BY location` resets the rank back to #1 for every new city.
- In the outer query, we simply filter `WHERE location_rank ≤ 3`.

3. SQL Query

```
WITH UserLocationEngagement AS (  
    SELECT  
        u.location,  
        u.user_id,  
        SUM(COALESCE(p.likes_cleaned, p.likes, 0) + p.shares + p.comments) AS total_engagement  
    FROM users u  
    JOIN posts p ON u.user_id = p.user_id  
    GROUP BY u.location, u.user_id  
)  
,  
RankedUsers AS (  
    SELECT  
        location,  
        user_id,  
        total_engagement,  
        DENSE_RANK() OVER (PARTITION BY location ORDER BY total_engagement DESC) AS location_rank  
    FROM UserLocationEngagement  
)  
SELECT  
    location,  
    user_id,  
    total_engagement,  
    location_rank  
FROM RankedUsers  
WHERE location_rank ≤ 3  
ORDER BY location, location_rank  
LIMIT 15;
```

4. Query Output (Sample of Top 3 per City)

location	user_id	total_engagement	location_rank
Barcelona, Spain	user_nfo3ih5u	79,421	1
Barcelona, Spain	user_5s9ifp0y	58,184	2
Barcelona, Spain	user_24wzfb8b	54,887	3
Beijing, China	user_ttlouvlq	51,516	1
Beijing, China	user_3jedn9am	49,321	2
Beijing, China	user_cdzp4vm2	48,019	3
Berlin, Germany	user_m5z8a3sg	56,480	1
Berlin, Germany	user_6e99lerx	55,057	2
Berlin, Germany	user_rtp2dykx	52,748	3
Chicago, USA	user_ujllj1n7	72,227	1
Chicago, USA	user_pr7bcf45	57,393	2
Chicago, USA	user_0v6j8da1	54,105	3

5. Human Explanation of the Result

This query produces an exact, city-by-city localized creator leaderboard. In Barcelona, `user_nfo3ih5u` dominates first place with 79.4k engagement, while in Chicago, `user_ujllj1n7` takes gold with 72.2k engagement.

H3 — Platform Performance Compared With Its Own Average

1. Plain-English Objective

Every platform has its own baseline. Find the breakout viral hits that generated **at least double ($\geq 2\times$) their platform's specific average engagement**.

2. The Thought Process & Data Traps

- Comparing a post on Twitter against the overall platform average would be unfair if Twitter had a much lower engagement ceiling than Instagram.
- We first compute each platform's own baseline average in CTE `PlatformAvg`.
- We then evaluate individual post engagement in CTE `PostStats` and join on `platform`.
- The condition `ps.post_engagement ≥ 2 * pa.platform_avg_engagement` isolates the true outliers.

3. SQL Query

```
WITH PlatformAvg AS (  
    SELECT  
        platform,  
        AVG(COALESCE(likes, 0) + shares + comments) AS platform_avg_engagement  
    FROM posts  
    WHERE platform IS NOT NULL  
    GROUP BY platform  
) ,  
PostStats AS (  
    SELECT  
        p.post_id,  
        p.platform,  
        (COALESCE(p.likes, 0) + p.shares + p.comments) AS post_engagement  
    FROM posts p  
    WHERE p.platform IS NOT NULL  
)  
SELECT  
    ps.post_id,  
    ps.platform,  
    ps.post_engagement,  
    ROUND(pa.platform_avg_engagement, 2) AS platform_avg_engagement  
FROM PostStats ps  
JOIN PlatformAvg pa ON ps.platform = pa.platform  
WHERE ps.post_engagement ≥ 2 * pa.platform_avg_engagement  
ORDER BY ps.post_engagement DESC  
LIMIT 10;
```

4. Query Output

post_id	platform	post_engagement	platform_avg_engagement
ycjj5zzt7mvx	Instagram	7,893	3,417.67
wo7py9al3g3t	Reddit	7,793	3,453.39
gmoeib832zbs	Facebook	7,764	3,456.55
5kvuyvf38nqx	YouTube	7,755	3,453.37
pvfl3d8hj7jd	Instagram	7,738	3,417.67
tdgjjylpua20	UNKNOWN	7,723	3,374.55
tne7s3o4l4wd	Instagram	7,712	3,417.67
a1kiwl618kzy	Facebook	7,683	3,456.55
fp89q1ickn9w	Twitter	7,628	3,345.93
5n161ir5hhhr	YouTube	7,610	3,453.37

5. Human Explanation of the Result

With platform baseline averages sitting between 3,345 and 3,456 interactions, any post achieving 6,900+ interactions qualifies as a $2\times$ statistical breakout. The top 10 posts achieved between 7,610 and 7,893 interactions—over **2.3 times** their respective platform norms!

H4 — Follower to Engagement Anomaly (Micro-Creators in the Top 10%)

1. Plain-English Objective

Uncover the "giant slayers": creators with tiny audiences (**fewer than 5,000 followers**) who nonetheless generated so much total engagement that they placed into the **top 10% (90th percentile)** of all creators on the entire platform.

2. The Thought Process & Data Traps

- How do you calculate a percentile rank in SQL?
- We use the advanced window function `PERCENT_RANK() OVER (ORDER BY total_engagement)` inside CTE `UserTotals`.
- `PERCENT_RANK()` outputs a value between 0.00 and 1.00 . A score ≥ 0.90 means the creator outperformed 90% of the user base.
- We filter for `follower_count < 5000 AND engagement_percentile  $\geq$  0.90`.

3. SQL Query

```
WITH UserTotals AS (  
  SELECT  
    u.user_id,  
    u.follower_count,  
    u.location,  
    SUM(COALESCE(p.likes_cleaned, p.likes, 0) + p.shares + p.comments) AS total_engagement,  
    PERCENT_RANK() OVER (ORDER BY SUM(COALESCE(p.likes_cleaned, p.likes, 0) + p.shares + p.comments))  
  FROM users u  
  JOIN posts p ON u.user_id = p.user_id  
  GROUP BY u.user_id, u.follower_count, u.location  
)  
SELECT  
  user_id,  
  follower_count,  
  location,  
  total_engagement,  
  ROUND(engagement_percentile * 100, 2) AS engagement_percentile_pct  
FROM UserTotals  
WHERE follower_count < 5000  
  AND engagement_percentile  $\geq$  0.90  
ORDER BY total_engagement DESC  
LIMIT 10;
```

4. Query Output

user_id	follower_count	location	total_engagement	engagement_percentile_pct
user_uerv85na	1,824	Rome, Italy	71,836	99.73%
user_hdasa0iaa	1,620	Rio de Janeiro, Brazil	66,500	99.47%
user_n0ok02rt	2,531	Dubai, UAE	63,716	98.87%
user_fgjkkrie	2,211	Lyon, France	62,690	98.80%
user_6wra58f7	4,953	Johannesburg, South Africa	55,557	96.73%
user_ogtvuuki	4,459	Cairo, Egypt	55,128	96.26%
user_rr1uzkql	1,069	Milan, Italy	54,817	96.06%
user_r7eg1rac	2,052	Houston, USA	54,552	95.86%
user_5oe5t3js	3,151	London, UK	53,170	94.86%
user_67hyf45u	898	Vancouver, Canada	51,406	93.20%

5. Human Explanation of the Result

Examine `user_67hyf45u` in Vancouver: with **only 898 followers**, this creator generated 51,406 interactions, placing them in the **93.2nd percentile** of all users! And `user_uerv85na` in Rome (1,824 followers) achieved 71.8k interactions, beating **99.73% of all creators** on the platform. This proves that algorithm-driven distribution rewards consistency and content over follower vanity metrics.

H5 — Identify Data Anomalies in Corrupted Dataset

1. Plain-English Objective

Scan the raw corrupted posts table (`posts_corrupted`) and build an audit report that classifies every corrupted record into its specific anomaly category: Negative Likes, Missing Platform, Missing Text Content, or HTML Tags/Entities.

2. The Thought Process & Data Traps

- A single post can suffer from multiple simultaneous corruptions (e.g. both a negative like count AND injected HTML markup).
- Using a series of `CASE` statements with `ELSE` would mask secondary corruptions because `CASE` stops evaluating at the first true condition.
- The correct architectural design is **UNION ALL** across four independent anomaly detectors:
 - `likes < 0` $\rightarrow$ Negative Likes
 - `platform IS NULL OR TRIM(platform) = ''` $\rightarrow$ Missing Platform
 - `text_content IS NULL OR TRIM(text_content) = ''` $\rightarrow$ Missing Text Content
 - `text_content LIKE '%&%' OR '%<div>%' ...` $\rightarrow$ HTML Entities or Tags in Text

3. SQL Query

```
SELECT post_id, 'Negative Likes' AS anomaly_type
FROM posts_corrupted
WHERE likes < 0

UNION ALL

SELECT post_id, 'Missing Platform' AS anomaly_type
FROM posts_corrupted
WHERE platform IS NULL OR TRIM(platform) = ''

UNION ALL

SELECT post_id, 'Missing Text Content' AS anomaly_type
FROM posts_corrupted
WHERE text_content IS NULL OR TRIM(text_content) = ''

UNION ALL

SELECT post_id, 'HTML Entities or Tags in Text' AS anomaly_type
FROM posts_corrupted
WHERE text_content LIKE '%&%;%'
   OR text_content LIKE '%<div>%'
   OR text_content LIKE '%</div>%'
   OR text_content LIKE '%<br>%'
   OR text_content LIKE '%<br />%'
   OR text_content LIKE '%<br />%'
ORDER BY post_id
LIMIT 15;
```

4. Query Output (Audit Log Sample)

post_id	anomaly_type
003s4u1m32tk	HTML Entities or Tags in Text
005g54tmt26m	Negative Likes
0066x8nnmouc	Missing Platform
0066x8nnmouc	HTML Entities or Tags in Text
00pk8aa72o8x	Missing Text Content
00u9otx16xfc	Missing Platform
014e8jqloj6h	Missing Text Content
01gbk9id4v75	Missing Text Content
01kgwhi645er	Negative Likes
01kgwhi645er	Missing Platform
02b0vyoya4hz	Missing Platform
02vdvvsogvsk	HTML Entities or Tags in Text

5. Human Explanation of the Result

Notice post `0066x8nnmouc` : it correctly appears **twice**—once for missing its platform header and once for having injected HTML markup. Similarly, `01kgwhi645er` was caught for both inverted likes and a dropped platform header. This output provides an exact checklist for automated cleaning pipelines.

H6 — Find the Most Suspicious High Impact Users

1. Plain-English Objective

Identify potentially manipulative accounts: creators with modest followings (**\$<\$ 10,000 followers**) who nonetheless maintain **above-average per-post engagement** AND have at least one post where **shares outpaced likes**.

2. The Thought Process & Data Traps

- We synthesize three conditions:
 1. Small account footprint (`u.follower_count < 10000`).
 2. High impact per post (`um.avg_engagement > om.overall_avg_engagement`).
 3. Evidence of viral skew (`SUM(CASE WHEN p.shares > p.likes THEN 1 ELSE 0 END) ≥ 1`).
- We construct two modular CTEs (`OverallMetric` for the platform mean and `UserMetrics` for user-level aggregations), filter in the outer query, and rank by total accumulated engagement using `DENSE_RANK()` .

3. SQL Query

```
WITH OverallMetric AS (  
    SELECT AVG(COALESCE(likes_cleaned, likes, 0) + shares + comments) AS overall_avg_engagement  
    FROM posts  
)  
,  
UserMetrics AS (  
    SELECT  
        u.user_id,  
        u.location,  
        u.follower_count,  
        COUNT(p.post_id) AS post_count,  
        AVG(COALESCE(likes_cleaned, likes, 0) + p.shares + p.comments) AS avg_engagement,  
        SUM(COALESCE(likes_cleaned, likes, 0) + p.shares + p.comments) AS total_engagement,  
        SUM(CASE WHEN p.shares > COALESCE(likes_cleaned, likes, 0) THEN 1 ELSE 0 END) AS suspicious_  
    FROM users u  
    JOIN posts p ON u.user_id = p.user_id  
    GROUP BY u.user_id, u.location, u.follower_count  
)  
SELECT  
    um.user_id,  
    um.location,  
    um.follower_count,  
    um.post_count,  
    ROUND(um.avg_engagement, 2) AS avg_engagement,  
    um.total_engagement,  
    DENSE_RANK() OVER (ORDER BY um.total_engagement DESC) AS rank_by_total_engagement  
FROM UserMetrics um  
CROSS JOIN OverallMetric om  
WHERE um.follower_count < 10000  
    AND um.avg_engagement > om.overall_avg_engagement  
    AND um.suspicious_share_posts ≥ 1  
ORDER BY um.total_engagement DESC  
LIMIT 10;
```

4. Query Output

user_id	location	follower_count	post_count	avg_engagement	total_engagement	rank
user_uerv85na	Rome, Italy	1,824	16	4,489.75	71,836	1
user_hdas0iauv	Rio de Janeiro, Brazil	1,620	16	4,156.25	66,500	2
user_68enpikx	Houston, USA	9,364	15	4,326.93	64,904	3
user_fgjkkrie	Lyon, France	2,211	14	4,477.86	62,690	4
user_9mtets0p	Johannesburg, South Africa	8,262	14	4,458.14	62,414	5
user_87kc1g6d	Houston, USA	7,238	14	4,333.57	60,670	6
user_g9g7onlw	Munich, Germany	7,981	13	4,490.15	58,372	7
user_cssz1rea	Tokyo, Japan	7,863	14	4,144.21	58,019	8
user_14ibkds2	Singapore	7,361	13	4,388.00	57,044	9
user_ogtvuuki	Cairo, Egypt	4,459	11	5,011.64	55,128	10

5. Human Explanation of the Result

user_uerv85na in Rome again ranks #1: with under 1,900 followers, they generated 71,836 total interactions with an incredible 4,489 average engagement per post, including posts where shares outnumbered likes.

user_ogtvuuki in Cairo achieved an astonishing **5,011 average engagement** per post across 11 posts. In a social platform audit, these creators are either viral grassroots sensations or using external amplification rings.

Part 4: Analytical Synthesis & Big Takeaways

Executing these 16 queries across Easy, Medium, and Hard tiers unlocks key macro insights regarding how the Social Engine operates:

1. The Fallacy of Follower Count (\$r \approx 0\$):

Queries **M2**, **H1**, and **H4** conclusively prove that follower count does not drive engagement. Micro-creators with \$<1,000\$ followers frequently pull 50,000+ interactions, rivaling macro-influencers with 49,000+ followers. The platform operates on a content-first recommendation algorithm.

2. Platform Specialization:

While total volumes are evenly balanced across networks (**E1**), each platform possesses a distinct behavioral signature (**E3**, **M4**):

- **Instagram** commands the highest average total engagement (4,043) and is the most lucrative channel for macro-creators.
- **Reddit** generates the highest comment density (511.2 comments/post), making it the primary hub for deep discussion and product critique.
- **Twitter** and **Facebook** generate the highest concentration of viral share-heavy anomalies (**E4**, **M5**).

3. Data Quality Integrity:

Queries **E4**, **H3**, and **H5** demonstrate why naive querying on raw data causes catastrophic reporting failures:

uncleaned negative likes would invert top rankings, while dropped headers would misattribute platform leadership. Utilizing clean, defensive SQL guarantees reproducible business intelligence.

Report compiled by Team Fernandoisfasterthanyou (pathakuntlabhaskarreddy@gmail.com) for the DataVortex Challenge.